NAME: BISMA FATIMA

CLASS: CS A

ROLL NO: F24605036

COURSE: OBJECT ORIENTED PROGRAMMING

COURSE CODE: CS-123

LECTURER: MISS TAYYABA KALSOOM

In LAB 12

TASK 01

```
//interface Shape
interface Shape {
    double area();
}
// Implement the interface
class Circle implements Shape {
    private double radius;
    // Constructor to initialize the radius
    public Circle(double radius) {
        this.radius = radius;
    }
    // Implement the area method
    @Override
    public double area() {
        return Math.PI * radius * radius;
    }
}
public class Main {
    public static void main(String[] args) {
        Circle circle = new Circle(5.0); // Creating a Circle object with radius 5.0
        System.out.println("Area of the circle: " + circle.area()); // Printing the area
    }
}
```

OUTPUT

```
Area of the circle: 78.53981633974483
```

TASK 02

```
// Define the interface Payable
interface Payable {
    double computePay();
}
// Implement the Payable interface
class Employee implements Payable {
    private double monthlySalary;
    private double bonus;
    private double overtimePay;
    // Constructor to initialize
    public Employee(double monthlySalary, double bonus, double overtimePay) {
        this.monthlySalary = monthlySalary;
        this.bonus = bonus;
```

```
        this.overtimePay = overtimePay;
    }
    // Implement the computePay method
    @Override
    public double computePay() {
        return monthlySalary + bonus + overtimePay;
    }
}
// Implement the Payable interface
class Contractor implements Payable {
    private double hourlyRate;
    private int hoursWorked;
    // Constructor to initialize
    public Contractor(double hourlyRate, int hoursWorked) {
        this.hourlyRate = hourlyRate;
        this.hoursWorked = hoursWorked;
    }
    // Implement the computePay method
    @Override
    public double computePay() {
        return hourlyRate * hoursWorked;
    }
}
//MAIN CLASS
public class TEST {
    public static void main(String[] args) {
        Employee employee = new Employee(5000, 1000, 500); // Fixed salary, bonus, overtime pay
        Contractor contractor = new Contractor(50, 160); // Hourly rate and hours worked

        System.out.println("Employee's Pay: " + employee.computePay());
        System.out.println("Contractor's Pay: " + contractor.computePay());
    }
}
```

OUTPUT

```
Employee's Pay: 6500.0
Contractor's Pay: 8000.0
```

TASK 03

```
// Appliance interface with basic methods
interface Appliance {
    void turnOn();
    void turnOff();
}
```

```
// Fan class implementing Appliance
class Fan implements Appliance {
    @Override
    public void turnOn() {
        System.out.println("Fan is ON.");
    }

    @Override
    public void turnOff() {
        System.out.println("Fan is OFF.");
    }
}
// AirConditioner class implementing Appliance
class AirConditioner implements Appliance {
    @Override
    public void turnOn() {
        System.out.println("Air Conditioner is ON.");
    }

    @Override
    public void turnOff() {
        System.out.println("Air Conditioner is OFF.");
    }
}
// Controller to manage appliances
class ApplianceController {
    public void control(Appliance appliance, boolean turnOn) {
        if (turnOn) {
            appliance.turnOn();
        } else {
            appliance.turnOff();
        }
    }
}
// Main class to test the system
public class ApplianceTest {
    public static void main(String[] args) {
        Appliance fan = new Fan();
        Appliance ac = new AirConditioner();
        ApplianceController controller = new ApplianceController();

        controller.control(fan, true);  // Turns the fan ON
        controller.control(ac, false);  // Turns the AC OFF
    }
}
```

OUTPUT

```
Fan is ON.
Air Conditioner is OFF.
```

TASK 04

```java
import java.util.Scanner;
import java.util.InputMismatchException;
public class DivisionHandler {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        try {
            // Taking input from the user
            System.out.print("Enter the first integer: ");
            int num1 = scanner.nextInt();
            System.out.print("Enter the second integer: ");
            int num2 = scanner.nextInt();
            // Performing division
            int result = num1 / num2;
            // Displaying the result
            System.out.println("Result: " + result);
        } catch (ArithmeticException e) {
            // Handling division by zero
            System.out.println("Error: Division by zero is not allowed.");
        } catch (InputMismatchException e) {
            // Handling invalid input
            System.out.println("Error: Please enter valid integers.");
        } finally {
            // Closing the scanner
            scanner.close();
            System.out.println("Program execution completed.");
        }
    }
}
```

OUTPUT

```
Enter the first integer: 5
Enter the second integer: 0
Error: Division by zero is not allowed.
Program execution completed.
```

TASK 05

```java
import java.util.ArrayList;
import java.util.List;
// Define the Device interface
interface Device {
    void start();
    void stop();
}
// Implement the Device interface for Fan
class Fan implements Device {
    @Override
    public void start() {
        System.out.println("Fan is now ON.");
    }
    @Override
    public void stop() {
        System.out.println("Fan is now OFF.");
    }
}
// Implement the Device interface for AirConditioner
class AirConditioner implements Device {
    @Override
    public void start() {
        System.out.println("Air Conditioner is now ON.");
    }
    @Override
    public void stop() {
        System.out.println("Air Conditioner is now OFF.");
    }
}
// Implement the Device interface for Television
class Television implements Device {
    @Override
    public void start() {
        System.out.println("Television is now ON.");
    }
    @Override
    public void stop() {
        System.out.println("Television is now OFF.");
```

```
    }
}
// Controller class to manage multiple devices using List<Device
class DeviceController {
    private List<Device> devices = new ArrayList<>();
    // Add devices to the list
    public void addDevice(Device device) {
        devices.add(device);
    }
    // Start all devices using a simple for loop
    public void startAllDevices() {
        for (int i = 0; i < devices.size(); i++) {
            devices.get(i).start();
        }
    }
    // Stop all devices using a simple for loop
    public void stopAllDevices() {
        for (int i = 0; i < devices.size(); i++) {
            devices.get(i).stop();
        }
    }
}
public class Devices {
    public static void main(String[] args) {
        // Create device instances
        Device fan = new Fan();
        Device ac = new AirConditioner();
        Device tv = new Television();
        // Create a controller and add devices
        DeviceController controller = new DeviceController();
        controller.addDevice(fan);
        controller.addDevice(ac);
        controller.addDevice(tv);
        // Start all devices
        System.out.println("Starting all devices:");
        controller.startAllDevices();
        // Stop all devices
        System.out.println("\nStopping all devices:");
        controller.stopAllDevices();
    }
}
```

OUTPUT

```
Starting all devices:
Fan is now ON.
Air Conditioner is now ON.
Television is now ON.

Stopping all devices:
Fan is now OFF.
Air Conditioner is now OFF.
Television is now OFF.
```

TASK 07

```java
import java.util.Scanner;
public class SimpleValidator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String name;
        int age;
        String email;
        // Validate Name
        System.out.print("Enter your name: ");
        name = scanner.nextLine();
        try {
            if (!name.matches("[A-Za-z ]+")) {
                System.out.println("Invalid Name: Only letters and spaces allowed.");
                name = "Invalid";
            }
        } catch (Exception e) {
            System.out.println("Error checking name.");
        }

        // Validate Age
        System.out.print("Enter your age: ");
        try {
            age = Integer.parseInt(scanner.nextLine());
            if (age <= 0 || age > 120) {
                System.out.println("Invalid Age: Must be between 1 and 120.");
                age = -1;
            }
        } catch (Exception e) {
            System.out.println("Invalid Age: Please enter a number.");
```

```
            age = -1;
        }

        // Validate Email
        System.out.print("Enter your email: ");
        email = scanner.nextLine();
        try {
            if (!email.contains("@") || !email.contains(".")) {
                System.out.println("Invalid Email: Email must contain '@' and a domain.");
                email = "Invalid";
            }
        } catch (Exception e) {
            System.out.println("Error checking email.");
        }

        // Display validated input
        System.out.println("\nUser Details:");
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Email: " + email);

        scanner.close();
    }
}
```

OUTPUT

```
Enter your name: BISMA FATIMA
Enter your age: 130
Invalid Age: Must be between 1 and 120.
Enter your email: BISMA@GMAIL.COM

User Details:
Name: BISMA FATIMA
Age: -1
Email: BISMA@GMAIL.COM
```

TASK 08

```
import java.util.Scanner;
class BankAccount {
    private double balance;
    public BankAccount(double initialBalance) {
```

```
        this.balance = initialBalance;
    }
    public double getBalance() {
        return balance;
    }
    // Withdraw method without using throw
    public void withdraw(double amount) {
        try {
            if (amount > balance) {
                System.out.println("Error: Insufficient funds! Available balance: " + balance);
                return; // Stop execution if funds are insufficient
            }
            balance -= amount;
            System.out.println("Withdrawal successful! New balance: " + balance);
        } catch (Exception e) {
            System.out.println("Unexpected error occurred.");
        }
    }
    // Retry transaction logic
    public void retryTransaction(double amount, int attempts) {
        while (attempts > 0) {
            System.out.println("Attempting transaction... Attempts left: " + attempts);
            withdraw(amount);
            if (amount <= balance) {
                break; // Exit loop if successful
            }
            attempts--;
        }
        if (attempts == 0) {
            System.out.println("Transaction failed after multiple attempts.");
        }
    }
}
// Main Class to Test the System
public class SimpleFinancialTransactionSystem {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        BankAccount account = new BankAccount(500); // Initial balance
        System.out.print("Enter withdrawal amount: ");
        double amount = scanner.nextDouble();
        account.withdraw(amount); // First attempt
        // Retry transaction logic if funds are insufficient
        account.retryTransaction(amount, 3);

        scanner.close();
    }
}
```

OUTPUT

```
Enter withdrawal amount: 678
Caught Exception: Error: Insufficient funds! Available balance: 500.0
```

TASK 09

```java
import java.util.Scanner;
public class ArrayExceptionHandling {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        // Define an array
        int[] numbers = {10, 20, 30, 40, 50};
        // Display array contents
        System.out.print("Array elements: ");
        for (int num : numbers) {
            System.out.print(num + " ");
        }
        System.out.println();
        // Ask user for an index
        System.out.print("Enter an index to access: ");
        int index = scanner.nextInt();
        try {
            // Attempt to access the user-specified index
            System.out.println("Value at index " + index + " : " + numbers[index]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Error: Invalid index! The array size is " + numbers.length);
        }
        scanner.close();
    }
}
```

OUTPUT

```
Array elements: 10 20 30 40 50
Enter an index to access: 6
Error: Invalid index! The array size is 5
```

TASK 10

```java
import java.util.Scanner;
public class AgeValidation {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
```

```
        try {
            System.out.print("Enter your age: ");
            int age = scanner.nextInt();
            // Validate age
            if (age < 18 || age == 0 || age > 90) {
                throw new Exception("Invalid Age! Age must be between 18 and 90.");
            }
            System.out.println("Valid Age: " + age);
        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        } finally {
            scanner.close(); // Ensure scanner is closed properly
        }
    }
}
```

OUTPUT

```
Enter your age: 79
Valid Age: 79
```